OS LAB - 24CS410P

COURSE OBJECTIVES:

Sl. No.	Course Objectives
1	Familiarize students with LINUX/UNIX OP and provide necessary skills for developing and debugging programs in these environments.
2	Learn shell script, creation and management of processes, IPC using shared memory and multithreads programing.
3	Analyze and develop process scheduling algorithms and process synchronization.

COURSE OUTCOMES (COs)

CO1	Implement shell programs and design process management and file system management with system calls.
CO2	Design and implement Inter Process Communication and multiple threads application.
CO3	Analyze and implement CPU scheduling algorithms and process synchronization.

Weeks	List of Programs
1	a) Write Shell Script to find the Fibonacci series. b) Implement the Priority (non preemptive) CPU scheduling algorithms by defining the process structure
2	a) Write Shell Script to perform an arithmetic operation using case b) Implement the Priority (Preemptive) CPU scheduling algorithms by defining the process structure
3	a) Write Shell Script to Check whether a given number is palindrome or not b) Implement FCFS scheduling algorithm by defining the process structure.
4	a) Write Shell Script to find largest of N numbers (storing numbers in an array) b) Implement the SJF (non preemptive) CPU scheduling algorithms by defining the process structure.
5	a) Write Shell Script to generate prime numbers. b) Implement the SJF (Preemptive) CPU scheduling algorithms by defining the process structure
6	a) Simulation of ls and rm commands using system calls b) Write a program to generate and print Prime numbers between a given range(between M & N) with the following requirements - M& N should be passed as command line arguments - Error checking should be done to verify the

	required number of arguments at the command line - Parent program should create a child and distribute the task of generating Prime numbers to its child. - The code for generating Prime numbers should reside in different program. - Child should write the generated Prime numbers to a shared memory. - Parent process has to print by retrieving the Prime numbers from the shared memory
7	a) Write a program to perform the following tasks using system calls: i. Parent process should create a child process ii. Both parent child processes should display their pid and iii. Load a new program into child process iv. The parent process should terminate after the child process terminates
8	a) Program to demonstrate the creation of Zombie and Orphan processes. b) Write a program with two threads and a main thread. Schedule the task of calculating the natural sum up to 'n' terms and factorial of 'n' on these threads. Note: The main thread as parameter to remaining threads. Terminate the threads using system calls.
9	a) Program to read from a file and write into a new file using I/O system calls for file I/O. b) Write a program to generate and print Fibonacci series with the following requirements: - Parent program should create a child and distribute the task of generating Fibonacci series to its child. - The code for generating Fibonacci series should reside in different program. - Child should write the generated Fibonacci series to a shared memory. - Parent process has to print by retrieving the Fibonacci series from the shared memory. Implement the above using shmget and shmat
10	a) Write a program to generate and print Fibonacci series with the following requirements: - Parent program should create a child and distribute the task of generating Fibonacci series to its child. - The code for generating Fibonacci series should reside in different program. - Child should write the generated Fibonacci series to a shared memory. - Parent process has to print by retrieving the Fibonacci series from the shared memory.
11	a) Implement Round Robin CPU scheduling algorithm with arrival time by defining the process structure.
12	Implement the following using mutex and semaphores: a. Producer Consumer problem b. Reader's writer's problem